# 05. Absolutní hodnota v lineárních funkcích, rovnicích, nerovnicích, grafy funkcí s absolutními hodnotami MO 27

pojem absolutní hodnota, definice význam

grafické a početní řešení rovnic a nerovnic s abs. hodnotou

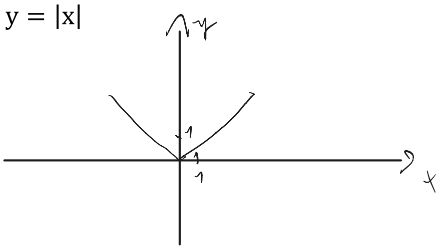
Teorie (+ tabulky):

Které příklady mi nešly:

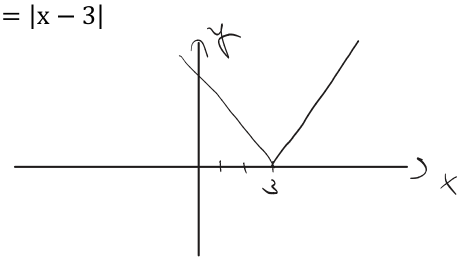
Co mi nejde? Dotazy?

1. Narýsujte grafy funkcí s D(f)=R:

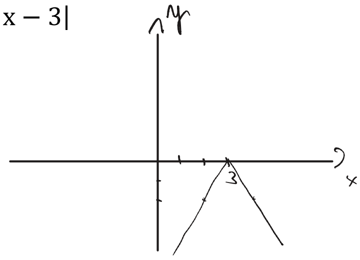
$f_1: y = |x|$



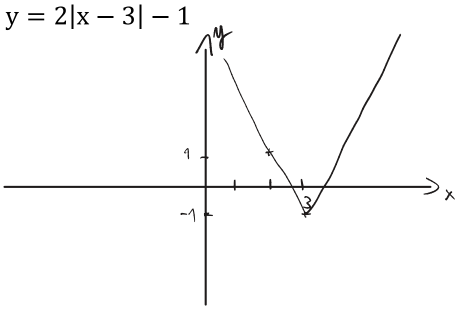
$f_2: y = |x - 3|$



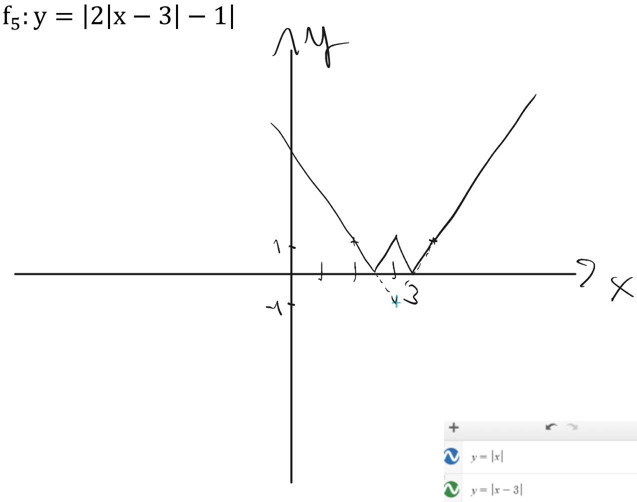
$f_3: y = -2|x - 3|$

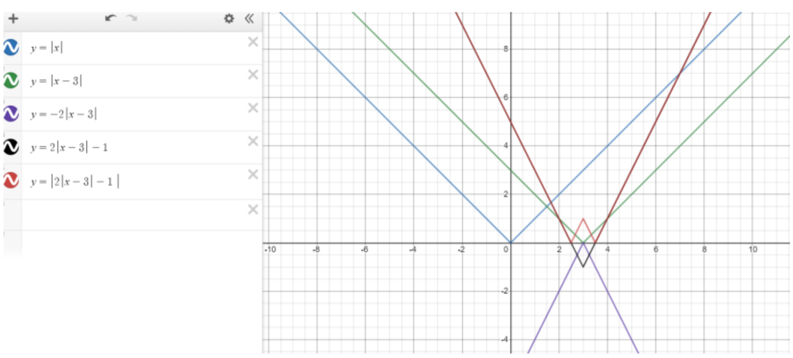


$f_4: y = 2|x - 3| - 1$



$f_5: y = |2|x - 3| - 1|$

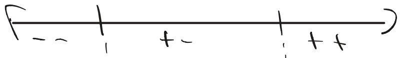


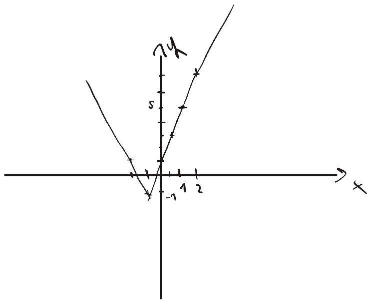


2. Narýsujte grafy funkcí s D(f)=R. Určete obor hodnot, souřadnice průsečíků s osami souřadnic, vyšetřete monotonii

$f_1: y = |3x+2| - |x-1|$

$-\frac{2}{3}$ $\quad 1$




$y = -3x-2+x-1$ | $3x+2+x-1 = 4x+1$ | $3x+2-x+1 = 2x+3$

$y = -2x-3$ | $y = 4x+1$ | $y = 2x+3$

| x | −1 | −2 |
|---|---|---|
| y | −1 | 1 |

| x | 0 | 0,5 |
|---|---|---|
| y | 1 | 3 |

| x | 1 | 2 |
|---|---|---|
| y | 5 | 7 |

$y = 4\left(-\frac{2}{3}\right)+1 = -\frac{5}{3}$

$0 = -2x-3$
$3 = -2x$
$x = -\frac{3}{2}$

$0 = 4x+1$
$-1 = 4x$
$x = -\frac{1}{4}$

$D = R;\ H\langle -\frac{5}{3}; \infty)$

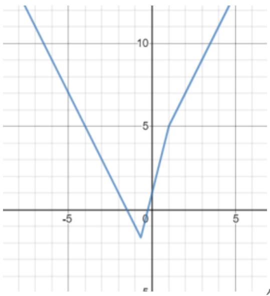
roste v $\langle -\frac{2}{3}; \infty)$

klesá v $(-\infty; -\frac{2}{3}\rangle$

$P_y[0;1]$

$P_{x_1}[-\frac{3}{2};0]$

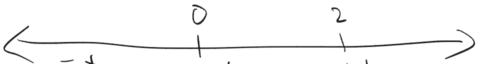
$P_{x_2}[-\frac{1}{4};0]$

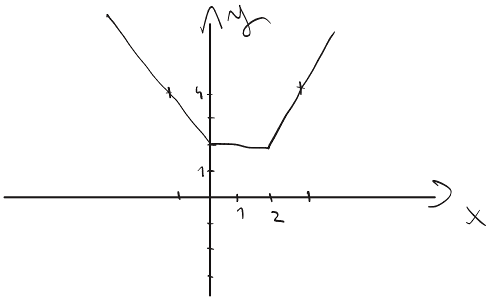
*f1:* 

, $H_f = \left\langle -\frac{5}{3}; \infty\right)$, *kles. v* $(-\infty; -\frac{2}{3}\rangle$, *rost. v* $\left\langle -\frac{2}{3}; \infty\right)$, $P_y[0;1]$, $P_{x1}\left[-\frac{3}{2};0\right]$, $P_{x2}\left[-\frac{1}{4};0\right]$ *]*

$f_2: y = |x| + |2-x|$

0 $\quad$ 2




$y = -x+2-x$ | $y = 2$ | $y = x-2+x$

$y = -2x+2$ | | $y = 2x-2$

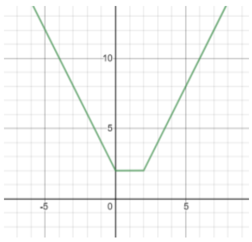
| x | −1 | −2 |
|---|---|---|
| y | 4 | 6 |

| x | 2 | 3 |
|---|---|---|
| y | 2 | 4 |

$P_x$ – není $\quad$ klesá v $(-\infty;0\rangle$; roste v $\langle 2;\infty)$

$P_y[0;2]$ $\quad$ konst. v $\langle 0;2\rangle$

$D = R;\ H = \langle 2;\infty)$

*[f2:* 

, $H_f = \langle 2;\infty)$, *kles. v* $(-\infty;0\rangle$, *konst. v* $\langle -0; 2\rangle$, *rost. v* $\langle 2;\infty)$; $P_y[0;2]$, $P_x$ *neex.]*

$4$

$f_3: y = 2|x-4| + x$

$4$

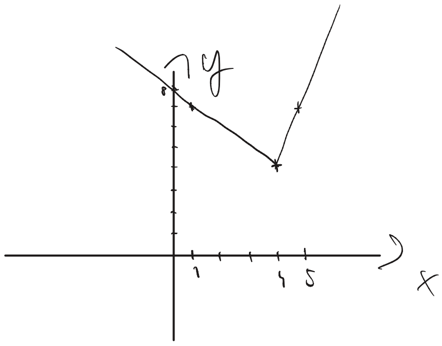
$-$ | $+$

$y = -2x+8+x$ | $y = 2x-8+x$

$y = -x+8$ | $y = 3x-8$

| x | 0 | 1 |
|---|---|---|
| y | 8 | 7 |

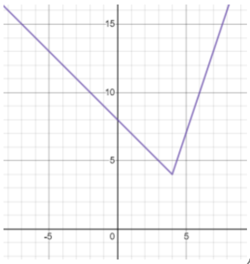
| x | 4 | 5 |
|---|---|---|
| y | 4 | 7 |




$P_y[0;8]$ $P_x$ – není!

klesá v $(-\infty;4)$; roste v $\langle 4;\infty)$

$D = \mathbb{R}$; $H = \langle 4;\infty)$

[$f_3$: 

, $H_f = \langle 4;\infty)$, kles. v $(-\infty;4)$, rost. v $\langle 4;\infty)$, $P_y[0; 8]$, $P_x$ neex.]

$-3$ $\quad$ $1$

$f_4: y = |x+3| - 2|1-x|$

$-3$ $\quad$ $1$

$- +$ | $+ +$ | $+ -$

$y = -x-3-2+2x$ | $y = x+3-2+2x$ | $y = x+3+2-2x$

$y = x-5$ | $y = 3x+1$ | $y = -x+5$

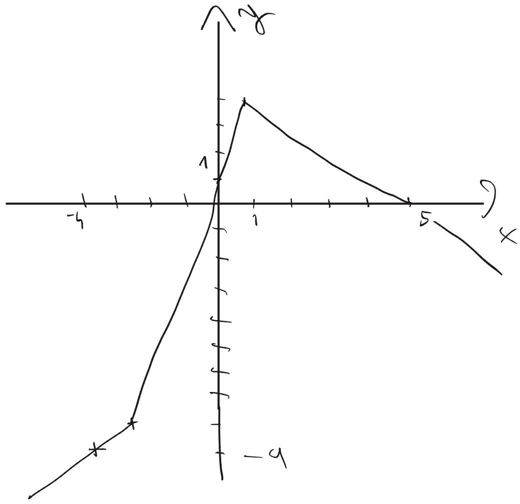
| x | -4 | -3 |
|---|---|---|
| y | -9 | -8 |

| x | -3 | 0 |
|---|---|---|
| y | -8 | 1 |

| x | 1 | 2 | 5 |
|---|---|---|---|
| y | 4 | 3 | 0 |

$0 = 3x+1$

$x = -\frac{1}{3}$

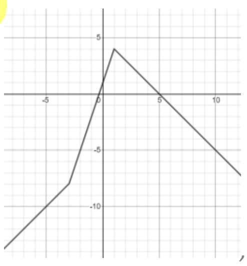



$P_{x_1}[-\frac{1}{3};0]$ $P_{x_2}[5;0]$

$P_y[0;1]$

roste v $(-\infty;1\rangle$; klesá $\langle 1;\infty)$

$D = \mathbb{R}$; $H = (-\infty;4\rangle$

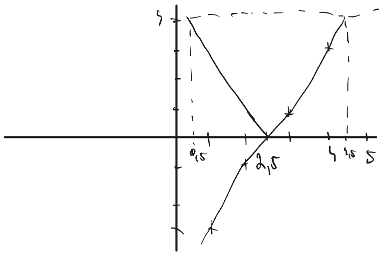
[$f_4$: 

, $H_f = (-\infty;4\rangle$, rost. v $(-\infty;1\rangle$, kles. v $\langle 1;\infty)$, $P_y[0; 1]$, $P_{x1}\left[-\frac{1}{3}; 0\right]$, $P_{x2}[5; 0]$ ]

3. Řešte graficky:

a) $|2x - 5| = 4$

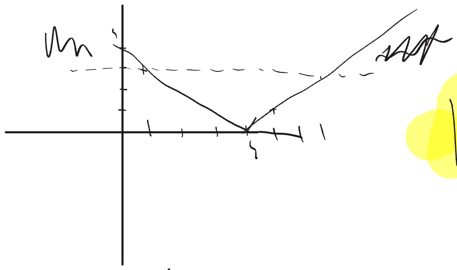
*$[P = \left\{\frac{1}{2}; \frac{9}{2}\right\}]$*



$K = \{0{,}5; 4{,}5\}$

b) $|4 - x| > 3$

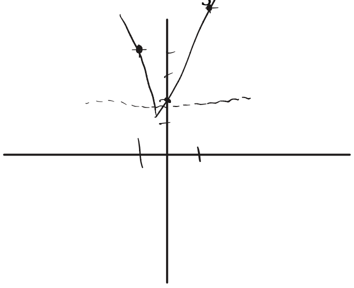
*$[P = (-\infty; 1) \cup (7; \infty)]$*



$K = (-\infty; 1) \cup (7; \infty)$

c) $|2 + 6x| \leq \frac{4}{3}$

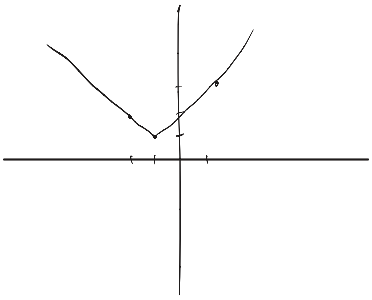
*$[P = \langle -\frac{5}{9}; -\frac{1}{9} \rangle]$*



$K = \langle -\frac{5}{9}; -\frac{1}{9} \rangle$

d) $|x + 1| + 1 = 0$

*$[P = \emptyset]$*

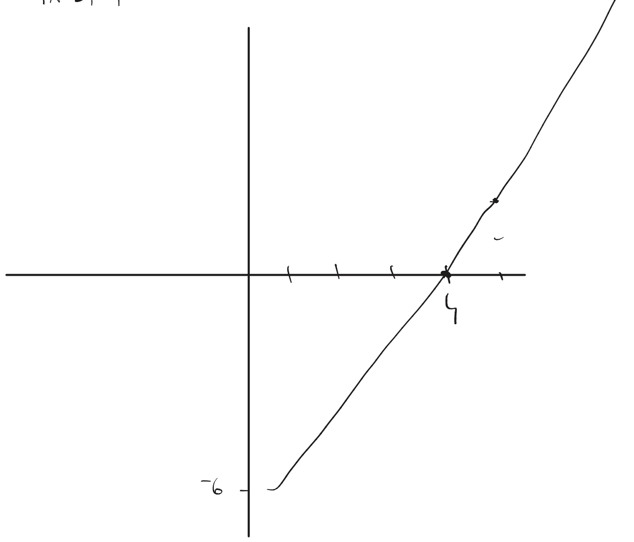


$K = \emptyset$

e) $|x + 3| = |11 - x|$

*$[P = \{4\}]$*

$|x+3| - |11 - x| = 0$



$K = \{4\}$

4. Řešte početně:

a) $|x-2| - |3x-1| = 2x-3$

$2 \qquad \frac{1}{3}$

$[P = \{1\}]$

$\frac{1}{3} \qquad 2$

$- - \qquad - + \qquad + +$

$-x+2+3x-1=2x-3$ | $-x+2-3x+1=2x-3$ | $x-2-3x+1=2x-3$

$1 \neq -3$ | $x=1$ | $x=\frac{1}{2}$

$\emptyset$ | | $\emptyset$ (není v intervalu)

$K=\{1\}$

$-\frac{4}{3} \qquad 1$

b) $|3x+4| + |x-1| \geq 12 - x$

$\left[P = (-\infty;-5\rangle \cup \left\langle\frac{9}{5};\infty\right)\right]$

$-\frac{4}{3} \qquad 1$

$- - \qquad + - \qquad + +$

$-3x-4-x+1 \geq 12-x$ | $3x+4-x+1 \geq 12-x$ | $3x+4+x-1 \geq 12-x$

$-4x-3 \geq 12-x$ | $2x+5 \geq 12-x$ | $4x+3 \geq 12-x$

$-3x \geq 15$ | $3x \geq 7$ | $5x \geq 9$

$x \leq -5$ | $x \geq \frac{7}{3}$ | $x \geq \frac{9}{5}$

mimo interval

$\emptyset$

$K=(-\infty;-5\rangle \cup \left\langle\frac{9}{5};\infty\right)$

c) $|1-2x| + |2+3x| < 11$

$\frac{1}{2} \qquad -\frac{2}{3}$

$\left[P = \left(-\frac{12}{5};2\right)\right]$

$-\frac{2}{3} \qquad \frac{1}{2}$

$+ - \qquad + + \qquad - +$

$1-2x-2-3x<11$ | $1-2x+2+3x<11$ | $-1+2x+2+3x<11$

$-5x-1<11$ | $x<8$ | $5x<10$

$x>-\frac{12}{5}$ | $\left\langle-\frac{2}{3};\frac{1}{2}\right)$ | $x<\frac{10}{5}$

| | $x<2$

$\left(-\frac{12}{5};-\frac{2}{3}\right)$ | | $\left\langle\frac{1}{2};2\right)$

$K=\left(-\frac{12}{5};2\right)$

d) $|5-x| - |x-3| = 2|x+1|$

$[P = \{-2; 0\}]$

$+ - -$

$5 - \cancel{x} + \cancel{x} - 3 = -2x - 2$

$4 = -2x$

$x = -2$

$+ - +$

$5 - \cancel{x} + \cancel{x} - 3 = 2x + 2$

$2x = 0$

$x = 0$

$+ + +$

$5 - x - x + 3 = 2x + 2$

$-4x = -6$

$x = \frac{3}{2}$

$\emptyset$ není v int.

$- + +$

$-5 + \cancel{x} - \cancel{x} + 3 = 2x + 2$

$2x = -4$

$x = -2$

$\emptyset$ není v int.

$K = \{-2; 0\}$

e) $|5-x| - |x-3| = 2(x+1)$

$[P = \{0\}]$

$+ -$

$5 - \cancel{x} + \cancel{x} - 3 = 2x + 2$

$x = 0$

$+ +$

$5 - x - x + 3 = 2x + 2$

$-4x = -6$

$x = \frac{3}{2}$

$\emptyset$

$- +$

$-5 + \cancel{x} - \cancel{x} + 3 = 2x + 2$

$-4 = 2x$

$x = -2$

$\emptyset$

$K = \{0\}$

f) $|x-3| + 3|x-1| = 2x + 1$

$\left[P = \left\{\frac{5}{6}; \frac{7}{2}\right\}\right]$

$- -$

$-x + 3 - 3x + 3 = 2x + 1$

$-6x = -5$

$x = \frac{5}{6}$

$- +$

$\cancel{-x} + \cancel{3} + \cancel{3x} - \cancel{3} = \cancel{2x} + 1$

$0 \neq 1$

$\emptyset$

$+ +$

$x - 3 + 3x - 3 = 2x + 1$

$2x = 7$

$x = \frac{7}{2}$

$K = \left\{\frac{5}{6}; \frac{7}{2}\right\}$

g) $|2x-3| \geq |3x-2|$ $\qquad [P = \langle -1; 1\rangle]$

$\frac{3}{2}$ $\quad \frac{2}{3}$

$\frac{2}{3}$ $\qquad \frac{3}{2}$

$- \, -$ $\qquad - \, +$ $\qquad + \, +$

$-2x+3 \geq -3x+2$

$x \geq -1$

$\langle -1; \frac{2}{3})$

$-2x+3 \geq 3x-2$

$-5x \geq -5$

$x \leq 1$

$\langle \frac{2}{3}; 1\rangle$

$2x-3 \geq 3x-2$

$-x \geq 1$

$x \leq -1$

$\emptyset$

$K = \langle -1; 1\rangle$

h) $|x+1| - |x-2| + |x-5| < 6$ $\qquad [P = (-4; 8) - \{2\}]$

$-1 \quad 2 \quad 5$

$-1$ $\qquad 2$ $\qquad 5$

$- - -$ $\qquad + - -$ $\qquad + + -$ $\qquad + + +$

$-x-1+x-2-x+5 < 6$

$-x < 4$

$x > -4$

$(-4; -1)$

$x+1+x-2-x+5 < 6$

$x < 2$

$\langle -1; 2)$

$x+1-x+2-x+5 < 6$

$-x < -2$

$x > 2$

$(2; 5\rangle$

$x+1-x+2+x-5 < 6$

$x < 8$

$\langle 5; 8)$

$K = (-4; 8) - \{2\}$

i) $3|x-1| + |3x-1| \leq x-1$ $\qquad [P = \emptyset]$

$1 \quad \frac{1}{3}$

$\frac{1}{3}$ $\qquad 1$

$- \, -$ $\qquad - \, +$ $\qquad + \, +$

$-3x+3-3x+1 \leq x-1$

$-7x \leq -5$

$x \geq \frac{5}{7}$

$\emptyset$

$-3x+3+3x-1 \leq x-1$

$3 \leq x$

$x \geq 3$

$\emptyset$

$3x-3+3x-1 \leq x-1$

$5x \leq 3$

$x \leq \frac{3}{5}$

$\emptyset$

$K = \emptyset$

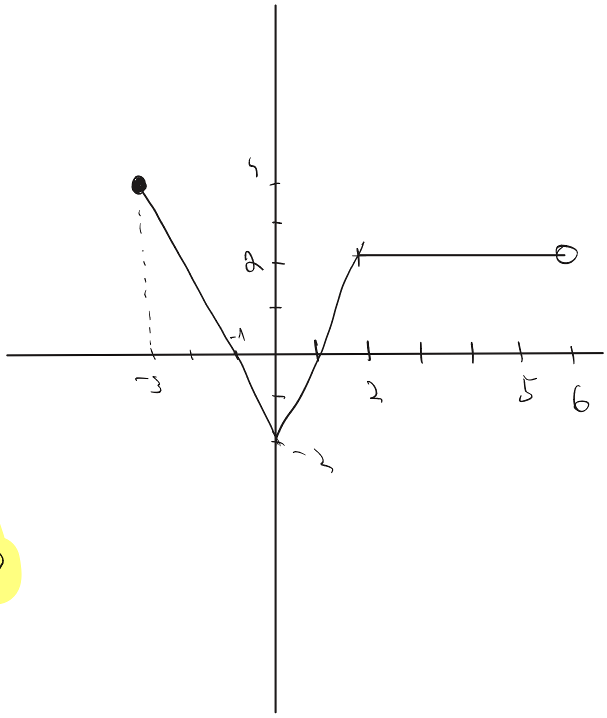
0 2

5. Narýsujte graf funkce, určete její obor hodnot: $f: y = |2x| - |x - 2| - x, D_f = \langle -3;6)$

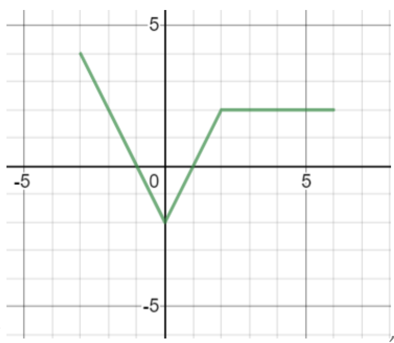

| $y = -2x + x - 2 - x$ | $y = 2x + x - 2 - x$ | $y = 2x - x + 2 - x$ |
|---|---|---|
| $y = -2x - 2$ | $y = 2x - 2$ | $y = 2$ |

| x | -1 | -2 | -3 |
|---|---|---|---|
| y | 0 | 2 | 4 |

| x | 0 | 1 |
|---|---|---|
| y | -2 | 0 |




$H = \langle -2;4\rangle$

[f: 

, $H_f = \langle -2; 4\rangle$]